

Premise: 1) $S(\text{Linda}) \wedge RC(\text{Linda})$ 2) $\forall x (RC(x) \Rightarrow T(x))$

Conclusion: $\exists x (S(x) \wedge T(x))$

Steps

Reason

1. $S(\text{Linda}) \wedge RC(\text{Linda})$ Premise
2. $\forall x (RC(x) \Rightarrow T(x))$ Premise
3. $RC(\text{Linda}) \Rightarrow T(\text{Linda})$ Universal instantiation on 2
4. $S(\text{Linda})$ Simplification on 1
5. $RC(\text{Linda}) \wedge S(\text{Linda})$ Commutative law on 1
6. $RC(\text{Linda})$ Simplification on 5
7. $T(\text{Linda})$ Modus ponens on 6, 3
8. $S(\text{Linda}) \wedge T(\text{Linda})$ Conjunction on 4, 7
9. $\exists x (S(x) \wedge T(x))$ Existential generalization on 8

b) Each of five roommates, Melissa, Aaron, Ralph, Venezia and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore all five roommates can take a course in algorithms next year.

Soln: $DM(x)$: x has taken a course in discrete mathematics
 $Al(x)$: x has taken a course in algorithms.

Domain: Set of all students

Premise: 1) $DM(\text{Melissa}) \wedge DM(\text{Aaron}) \wedge DM(\text{Ralph}) \wedge DM(\text{Venezia}) \wedge DM(\text{Keeshawn})$ 2) $\forall x (DM(x) \Rightarrow Al(x))$

Conclusion: $Al(\text{Melissa}) \wedge Al(\text{Aaron}) \wedge Al(\text{Ralph}) \wedge Al(\text{Venezia}) \wedge Al(\text{Keeshawn})$

Steps

Reason

1. $DM(\text{Melissa}) \wedge DM(\text{Aaron}) \wedge DM(\text{Ralph}) \wedge DM(\text{Venezia}) \wedge DM(\text{Keeshawn})$ Premise